

Exercise 1.4

Compute the products in Exercises 1-4 using (a) the definition, as in Example 1, and (b) the row-vector rule for computing Ax . If a product is undefined, explain why.

1. $\begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 7 \end{bmatrix}$ $A = 3 \times 2$ matrix $x = (3 \times 1)$ vector
 $m=3, n=2$ x in $\mathbb{R}^3$
 The no. of columns of A does not match with the no. of rows of x . $\therefore$ The product Ax is undefined.

3. $\begin{bmatrix} 6 & 5 \\ -4 & -3 \\ 7 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \end{bmatrix} = 2 \begin{bmatrix} 6 \\ -4 \\ 7 \end{bmatrix} + (-3) \begin{bmatrix} 5 \\ -3 \\ 6 \end{bmatrix}$ (From the definition of Ax)

$= \begin{bmatrix} 2 \cdot 6 \\ 2 \cdot (-4) \\ 2 \cdot 7 \end{bmatrix} + \begin{bmatrix} (-3) \cdot 5 \\ (-3) \cdot (-3) \\ (-3) \cdot 6 \end{bmatrix}$ (Scalar multiplication)

$= \begin{bmatrix} 12 \\ -8 \\ 14 \end{bmatrix} + \begin{bmatrix} -15 \\ 9 \\ -18 \end{bmatrix} = \begin{bmatrix} 12+(-15) \\ -8+9 \\ 14+(-18) \end{bmatrix}$ (Vector addition)

$= \begin{bmatrix} -3 \\ 1 \\ -4 \end{bmatrix}$

$\begin{bmatrix} 6 & 5 \\ -4 & -3 \\ 7 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 6 \cdot 2 + 5 \cdot (-3) \\ (-4) \cdot 2 + (-3) \cdot (-3) \\ 7 \cdot 2 + 6 \cdot (-3) \end{bmatrix}$ (Row-vector rule for computing Ax)

$= \begin{bmatrix} -3 \\ 1 \\ -4 \end{bmatrix}$

In Exercises 5-8, use the definition of Ax to write the matrix equation as a vector equation, or vice-versa.